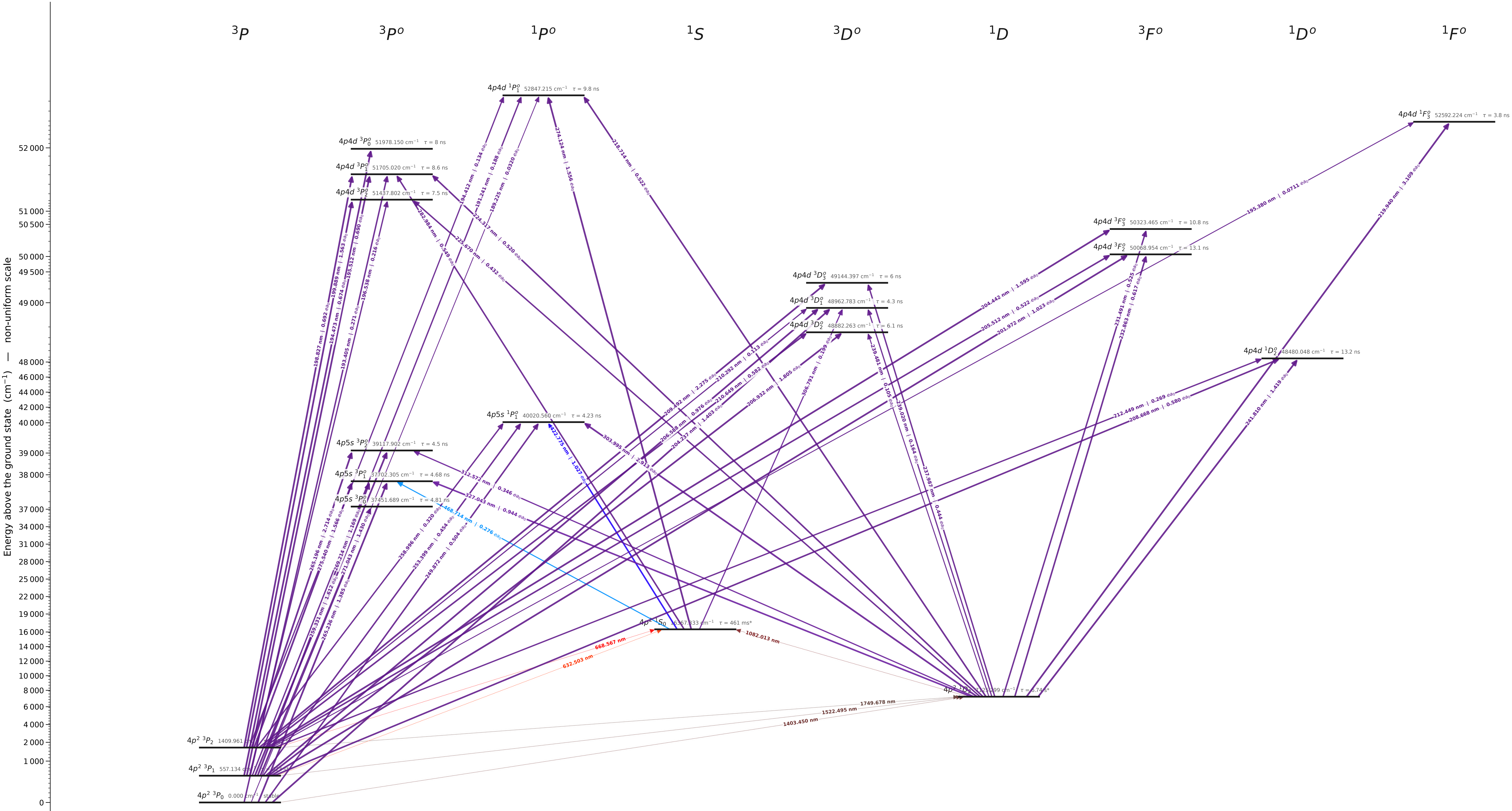
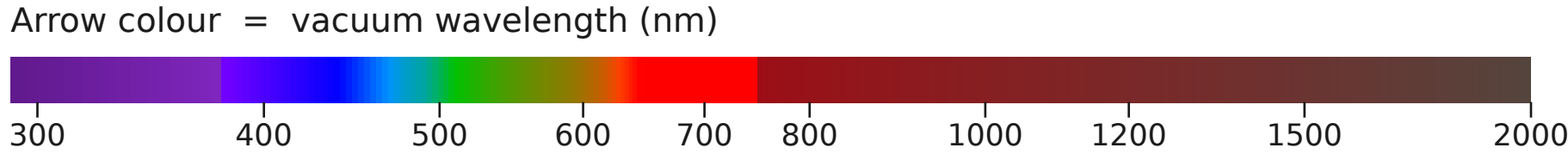


Germanium

Ge (Ge I). Strongest classified transitions below 2 μm (NIST): 20 levels, 54 lines.

Ge⁺ ionisation limit 63713.24 cm⁻¹ = 7.89943 eV (156.953 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle U || e r || U \rangle|$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J er J'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
4p ² ³ P ₁ - 4p5s ³ P ₁ ^o	269.2140	269.1341	1113.5842	1.169	4.730e+07	34.01	22.1	269.13 nm line used for collinear laser spectroscopy of G...
4p ² ³ P ₁ - 4p ² ¹ S ₀	632.5031	632.3282	473.9778	-	1.000e+00	3.452e-07	46.1	forbidden line within the ground configuration
4p ² ³ P ₂ - 4p ² ¹ S ₀	668.5666	668.3821	448.4107	-	6.800e-02	3.452e-07	3.13	forbidden line within the ground configuration
4p ² ¹ D ₂ - 4p ² ¹ S ₀	1082.0129	1081.7165	277.0692	-	1.100e+00	3.452e-07	50.7	forbidden line within the ground configuration
4p ² ³ P ₀ - 4p ² ¹ D ₂	1403.4499	1403.0663	213.6111	-	3.000e-05	2.361e-08	0.02	forbidden line within the ground configuration
4p ² ³ P ₁ - 4p ² ¹ D ₂	1522.4953	1522.0793	196.9086	-	5.030e-02	2.361e-08	33.9	forbidden line within the ground configuration
4p ² ³ P ₂ - 4p ² ¹ D ₂	1749.6778	1749.2001	171.3415	-	9.800e-02	2.361e-08	66	forbidden line within the ground configuration

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

⁷³Ge hyperfine structure ($I = 9/2$)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
4p ² ³ P ₁	15.5480(18)	-54.566(9)	F=7/2: -110.52 F=9/2: +20.83 F=11/2: +56.32
4p ² ³ P ₂	-64.4270(7)	111.825(13)	F=5/2: +759.95 F=7/2: +485.53 F=9/2: +160.66 F=11/2: -193.68 F=13/2: -551.89
4p5s ³ P ₁ ^o	-262.2(12)	40(6)	F=7/2: +1460.43 F=9/2: +235.53 F=11/2: -1169.90

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
4p ² ³ P ₁ - 4p5s ³ P ₁ ^o	269.214	70-72	-35.0(20)
4p ² ³ P ₁ - 4p5s ³ P ₁ ^o	269.214	73-72	8(4)
4p ² ³ P ₁ - 4p5s ³ P ₁ ^o	269.214	74-72	52.0(20)

Shift = ν(first isotope) - ν(second isotope).